HOMEWORK E.1 · EOG REVIEW · NC.8.NS.1 · NC.8.EE.1 · NC.8.G.9 · NC.8.EE.7 · NC.8.NS.2 · NC.8.F.4 · NC.8.G.6

EOG Review: Real Number System & exponent rules

Targets: NS.1-3, EE.1-2, G.9-2, EE.7-3, NS.2-3, NS.1-5, EE.1-6, F.4-5, EE.1-8, G.6-9

Name _____ Date _____ Block _____

Remember: Rational: the decimal stops or repeats. 9's rule: $0.\overline{45} = \frac{45}{99}$. **Irrational**: never stops, never repeats ($\sqrt{7}$, π). **Exponents** (same base): multiply → **add**; divide → **subtract**; power of a power → **multiply**; $a^0 = 1$; $a^{-n} = \frac{1}{a^n}$.

Example: $x^3 \cdot x^{-5} = x^{-2} = \frac{1}{x^2}$.

SCORE 3.0 Approaching · the basics

Targets: NS.1-3, EE.1-2, G.9-2, EE.7-3, NS.2-3

1 Which numbers are **rational**? NS.1-3

☐ $\sqrt{25}$ ☐ $\sqrt{12}$ ☐ $0.\overline{8}$ ☐ π

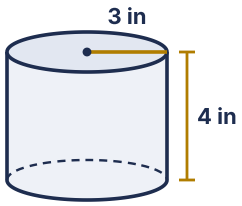
☐ -3.5 ☐ $\frac{2}{7}$

Select all that apply.

2 Simplify: $\frac{6^8}{6^3}$ EE.1-2

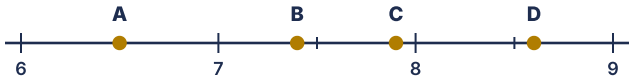
Ⓐ 6^5 Ⓑ 6^{11} Ⓒ 1^5 Ⓓ 6^{24}

3 A cylinder has radius 3 in and height 4 in. Find its volume in terms of π. REVIEW 9.2 G.9-2



4 Solve: $3(x+2) = 5x - 8$ REVIEW 3.4 EE.7-3

5 Which point is closest to $\sqrt{55}$? NS.2-3



Ⓐ A Ⓑ B Ⓒ C Ⓓ D

SCORE 4.0 Proficient · on target

Targets: NS.1-5, EE.1-6, F.4-5

NS.1-5

6 Use the 9's rule. Write $0.\overline{27}$ as a fraction in simplest form.

EE.1-6

7 Simplify: $\dfrac{12x^5y^{-2}}{4x^{-1}y^3}$

(A) $\dfrac{3x^6}{y^5}$

(B) $\dfrac{3x^4}{y}$

(C) $\dfrac{8x^6}{y^5}$

(D) $3x^4y$

REVIEW 6.5 F.4-5

8 Find the slope of the line through $(2, 5)$ and $(6, -3)$.

Hint: slope = $\dfrac{y_2 - y_1}{x_2 - x_1}$

SCORE 5.0 Mastery · go deeper

Targets: EE.1-8, G.6-9

EE.1-8

9 Explain why $2^{-3} = \dfrac{1}{8}$ and not -8. Use the pattern $2^2 = 4$, $2^1 = 2$, $2^0 = 1$.

Each time the exponent goes down by 1, _____ . So 2^{-3} = _____ . A negative exponent means _____ .

REVIEW 5.3 G.6-9

10 Sam says the Pythagorean Theorem and its converse are the same thing. Explain how they are different.

The theorem is used when _____ . The converse is used when _____ .